

Many end-users simply don't find this compelling, especially at \$50 per month. Unlike the open WiFi market, there is no room for innovators to roll out new service offerings or better technology, because everything must go through the network owner.

As noted above, standard WiFi technology provides only short-range connections, to a range of approximately 300 feet. This is insufficient for most residential broadband deployments. To deliver broadband to a home, the home must connect to a high-speed Internet trunk, which can be shared among many customers. Having a fast WiFi connection in a house doesn't substitute for DSL or a cable modem, because the wired connection is still necessary to reach the public Internet.

Despite these limitations, there are several approaches that could allow unlicensed wireless devices to deliver last-mile broadband service. Companies such as Nokia (with its Rooftop system), MeshNetworks and SkyPilot have created systems that use a meshed architecture. In other words, rather than connecting to a central hub, each device can send information to every other device it can see. Information can be routed through the network using many different paths, depending on capacity, line of sight, and other characteristics. The mesh approach gets around limitations that hobbled previous fixed-wireless systems in the last mile. Other companies such as Etherlinx and Motorola have created proprietary technologies on top of WiFi radios to allow significantly increased range in traditional point-to-multi-point deployments. Motorola claims its Canopy technology can serve up to 1,200 subscribers from a single access point at a range of up to two miles, operating in the unlicensed 5 GHz band. Unlicensed wireless connections could also serve as "tails" at the end of existing phone, cable, or fiber infrastructure in residential neighborhoods.

All these configurations have their limitations. As with any wireless service, connection quality depends on physical geography and the local spectral environment. As a result, it's